DELHI TECHNOLOGICAL UNIVERSITY

Predicting the Price of Used Cars using Machine Learning Techniques

AREA: DATA MINING

SUBMITTED TO:

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ACKNOWLEDGEMENT

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Secondly, We would also like to thank my parents and friends who helped us a lot in finalizing this project within the limited time frame.

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CERTIFICATE

This is to certify that Yanshul Narotra and Yash Nishal, students of B.Tech 2nd year 2020-2021 of this institution. This project report Titled "**Predicting the Price of Used Cars using Machine Learning Techniques**" has been prepared by them in partial fulfilment for the requirement to qualify the semester, to be submitted to DTU(Delhi Technological University), under the supervision and guidance of Mr MANOJ SETHI, Lecturer, Department of Computer Science.

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Introduction

Predicting the price of used cars in both an important and interesting problem. According to data obtained from the National Transport Authority , the number of cars registered between 2003 and 2013 has witnessed a spectacular increase of 234%. From 68, 524 cars registered in 2003, this number has now reached 160, 701. With difficult economic conditions, it is likely that sales of second-hand imported (reconditioned) cars and used cars will increase. It is reported that the sales of new cars has registered a decrease of 8% in 2013.

Meanwhile, demand for used cars has increased rapidly compared to pre-Covid times, and this mismatch between demand and supply has led to a 2-7 per cent increase in the price of used cars, said experts.

Predicting the resale value of a car is not a simple task. It is trite knowledge that the value of used cars depends on a number of factors. The most important ones are usually the age of the car, its make (and model), the origin of the car (the original country of the manufacturer), its mileage (the number of kilometers it has run) and its horsepower.

We will be predicting the costs of used cars given the data collected from various sources and distributed across various locations in India using different features such as:

Name: The brand and model of the car.
Location: The location in which the car is being sold or is available for purchase.
Year: The year or edition of the model.
Kilometers_Driven: The total kilometres driven in the car by the previous owner(s) in KM.
Fuel_Type: The type of fuel used by the car.
Transmission: The type of transmission used by the car.
Owner_Type: Whether the ownership is First-hand, Second hand or other.
Mileage: The standard mileage offered by the car company in kmpl or km/kg
Engine: The displacement volume of the engine in cc.
Power: The maximum power of the engine in bhp.
Seats: The number of seats in the car.
Price: The price of the used car in INR Lakhs.

Related work

Özer Çelik used Linear Reggresion. The main features used in this study were: mileage, manufacturer and estimate useful life. In the estimation model, the data quality and reliability are increased by using the actual sales prices of the vehicles.

Zhang et al. use Kaggle data-set to perform price prediction of a used car. The author evaluates the performance of several classification methods (logistic regression, SVM, decision tree, Extra Trees, AdaBoost, random forest) to assess the performance. Among all these models, random forest classifier proves to perform the best for their prediction task. This work uses five features (brand, powerPS, kilometer, sellingTime, VehicleAge) to perform the classification task after removal of irrelevant features and outliers from the dataset which gives an accuracy of 83.08% on the test data. We also use Kaggle data-set to perform prediction of used-car prices. However, the difference lies in the inclusion of few more relevant features in prediction model - the price of the car, and vehicleType. These two features play an important role in predicting the price of a used car which seems to be given less importance in the paper.

The report by Sameerchand Pudaruth, reviews the four most popular classification methods (Multiple Linear Regression Analysis, K-Nearest Neighbours, Decision Trees, Naïve Bayes) to perform prediction.

Nabarun Pal used Random Forest, a controlled learning method to estimate the price of used cars. The model was chosen after careful exploration data analysis to determine the effect of each feature on the price. A Random Forest with 500 Decision Trees was created to train the data. From the test results, it was found that the accuracy of the training was 95.82% and the test accuracy was 83.63%. The model was able to accurately estimate the price of the cars by choosing the most suitable features

Proposed work

In this Project report, we propose a Machine Learning model namely random forest Regressor to predict the prices of used cars given the features. The price is estimated based on the number of features as mentioned above. The intricate details about this model on the used car's data set along with the accuracy are narrated in-depth in the Result and Conclusion section. We then deploy Jupyter Notebook to display our results which are capable of predicting the price of a car given so many features of it. This deployed service is a result of our work, and it incorporates the data, ML model with the features. To summarize,

- First, we collect the data about used cars , identify important features that reflect the price

- Second, we preprocess and remove entries with NA values. Discard features that are not relevant for the prediction of the price.

- Third, we apply various models on the preprocessed dataset with features as inputs and the price as output and compared their accuracies in which random forest regressor came on top.

- Finally, we deploy Jupyter Notebook which incorporates all the features of the used cars and the random forest model to predict the price of a car.

Implementation

We used python libraries such as Pandas, Numpy, Matplotlib, Seaborn for graphical Analysis and Data preprocessing.

Data Pre-Processing

- We will remove all the duplicate rows from the data.

```
In [18]: data.duplicated().sum()
Out[18]: 0
```

(NO DUPLICATED DATA PRESENT)

- We will remove all the rows having missing or null data.

```
In [15]: data.isnull().sum()
Out[15]: Name                 0
         Location             0
         Year                 0
         Kilometers_Driven    0
         Fuel_Type            0
         Transmission         0
         Owner_Type           0
         Mileage              2
         Engine              36
         Power               36
         Seats               42
         Price                0
         dtype: int64
```

```
In [11]: data=data.dropna()
In [12]: data.isnull().sum()
Out[12]: Name                 0
         Location             0
         Year                 0
         Kilometers_Driven    0
         Fuel_Type            0
         Transmission         0
         Owner_Type           0
         Mileage              0
         Engine               0
         Power                0
         Seats                0
         Price                0
         dtype: int64
```

2 + 36 + 36 + 42 = 116 null values

- We will make Mileage, Engine, Power into purely float values by removing
  - “Kmpl” from Mileage
  - “CC” from Engine

- “bhp” from Power

- We will subtract 2000 from the Year feature to make the analysis simple
- We will split the Name column into the Brand and Model column

	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	Price
0	Maruti Wagon R LXI CNG	Mumbai	2010	72000	CNG	Manual	First	26.6 km/kg	998 CC	58.16 bhp	5.0	1.75
1	Hyundai Creta 1.6 CRDi SX Option	Pune	2015	41000	Diesel	Manual	First	19.67 kmpl	1582 CC	126.2 bhp	5.0	12.50
2	Honda Jazz V	Chennai	2011	46000	Petrol	Manual	First	18.2 kmpl	1199 CC	88.7 bhp	5.0	4.50
3	Maruti Ertiga VDI	Chennai	2012	87000	Diesel	Manual	First	20.77 kmpl	1248 CC	88.76 bhp	7.0	6.00
4	Audi A4 New 2.0 TDI Multitronic	Coimbatore	2013	40670	Diesel	Automatic	Second	15.2 kmpl	1968 CC	140.8 bhp	5.0	17.74

```
In [16]: temp = data["Name"].str.split(" ", n = 1, expand = True)
         data['Brand']=temp[0]
         data['Model']=temp[1]
         data['Engine']=data.Engine.str.split(" ").str[0]
         data['Power']=data.Power.str.split(" ").str[0]
         data['Mileage']=data.Mileage.str.split(" ").str[0]
         data['Year']=np.subtract(2020,data['Year'])
         data.head()
```

	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	Price	Brand	Model
0	Maruti Wagon R LXI CNG	Mumbai	10	72000	CNG	Manual	First	26.6	998	58.16	5.0	1.75	Maruti	Wagon R LXI CNG
1	Hyundai Creta 1.6 CRDi SX Option	Pune	5	41000	Diesel	Manual	First	19.67	1582	126.2	5.0	12.50	Hyundai	Creta 1.6 CRDi SX Option
2	Honda Jazz V	Chennai	9	46000	Petrol	Manual	First	18.2	1199	88.7	5.0	4.50	Honda	Jazz V
3	Maruti Ertiga VDI	Chennai	8	87000	Diesel	Manual	First	20.77	1248	88.76	7.0	6.00	Maruti	Ertiga VDI
4	Audi A4 New 2.0 TDI Multitronic	Coimbatore	7	40670	Diesel	Automatic	Second	15.2	1968	140.8	5.0	17.74	Audi	A4 New 2.0 TDI Multitronic

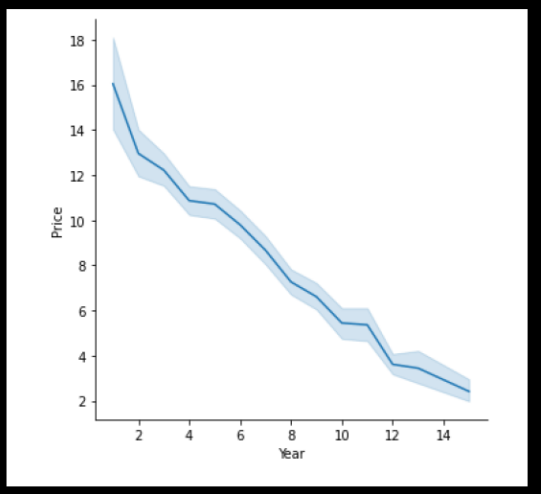
```
data.dropna(how="all",inplace=True)
data['Power']=data['Power'].astype(float)
data['Mileage']=data['Mileage'].astype(float)
data['Engine']=data['Engine'].astype(float)

data.dtypes
```

```
Location              object
Year                 float64
Kilometers_Driven    float64
Fuel_Type             object
Transmission          object
Owner_Type            object
Mileage              float64
Engine               float64
Power                float64
Seats                float64
Price                float64
Brand                 object
Model                 object
dtype: object
```

Graphical Analysis

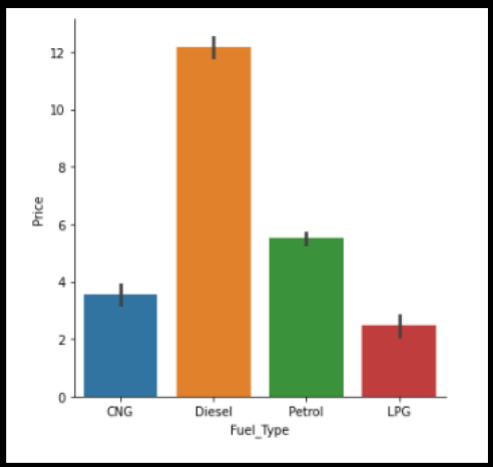
Though the curve is not perfectly linear, it can be seen that when the car is new (bought in recent years), it has a higher market value.



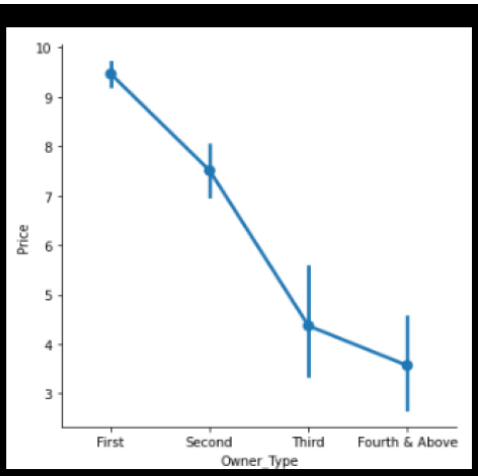

```
sns.relplot(x="Year", y="Price", kind="line", data=data)
```

The above graph clearly states the fact that diesel cars are more preferred than petrol cars. Thus, price of used cars follows the order: diesel>petrol>CNG>LPG .
This might be because:

- Petrol and Diesel have high engine capacity.
- Though CNG and LPG gives higher mileage, they are not safer and require high maintenance.
- Cost of petrol is high and diesel index is little complex and thus a little costlier than petrol engine.

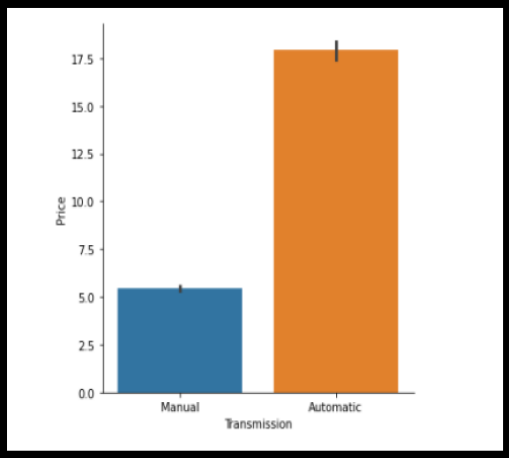



```
a=sns.catplot(x="Fuel_Type", y="Price", kind="bar", data=data)
```



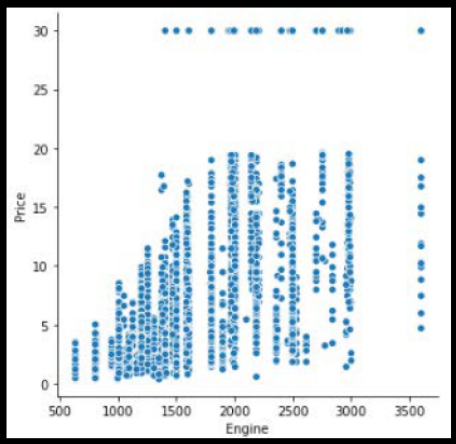

```
a=sns.catplot(x="Owner_Type", y="Price", kind="point",
  data=data,order=['First','Second','Third','Fourth & Above'])
a
```

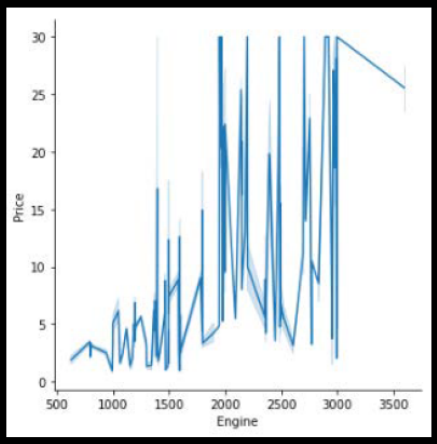
clearly, new car is having higher price than second hand car




```
sns.catplot(x="Transmission", y="Price", kind="bar", data=data)
```

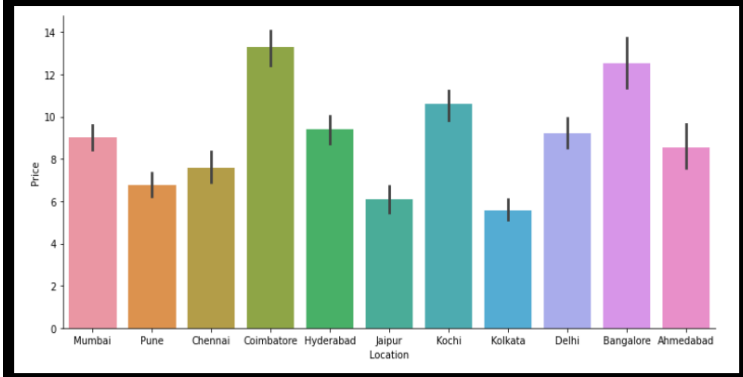
Price of automated transmission is higher than manual, again for obvious reasons



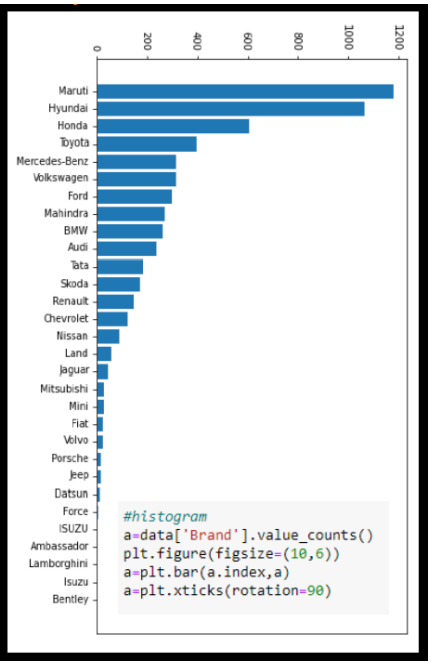



```
sns.relplot(x='Engine',y='Price',data=data,kind='scatter')
sns.relplot(x='Engine',y='Price',data=data,kind='line')
```

The dots are not perfectly forming a pattern, but more or less, it seems more the capacity of the engine, more price of the car.



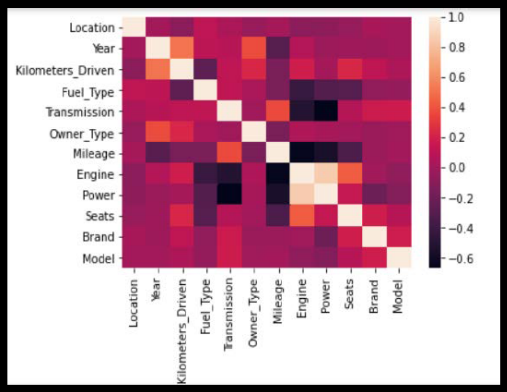

```
a=sns.catplot(x="Location", y="Price", kind="bar", data=data)
a.fig.set_figwidth(12)
a.fig.set_figheight(5)
```




```
#histogram
a=data['Brand'].value_counts()
plt.figure(figsize=(10,6))
a=plt.bar(a.index,a)
a=plt.xticks(rotation=90)
```

```
data.corr()
```

	Year	Kilometers_Driven	Mileage	Engine	Power	Seats	Price
Year	1.000000	0.512684	-0.287826	0.067942	-0.019287	-0.006873	-0.325440
Kilometers_Driven	0.512684	1.000000	-0.147014	0.188482	0.019695	0.232616	-0.171960
Mileage	-0.287826	-0.147014	1.000000	-0.644034	-0.552527	-0.330161	-0.363456
Engine	0.067942	0.188482	-0.644034	1.000000	0.860122	0.426556	0.678437
Power	-0.019287	0.019695	-0.552527	0.860122	1.000000	0.140161	0.795380
Seats	-0.006873	0.232616	-0.330161	0.426556	0.140161	1.000000	0.106004
Price	-0.325440	-0.171960	-0.363456	0.678437	0.795380	0.106004	1.000000




```
sns.heatmap(data.corr())
```

HEAT MAP SHOWING RELATION BETWEEN DIFFERENT FEATURES

Model Implementation

First, we will convert all the named features into numbers so that we can implement them into our model.

FOR EXAMPLE :

Owner_Type	First	Second	Third	Fourth & Above values
Owner_Type	0.00	1.00	2.00	3.00

	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	Price	Brand	Model
0	9.0	10.0	72000.0	0.0	1.0	0.0	26.60	998.0	58.16	5.0	1.75	18.0	1579.0
1	10.0	5.0	41000.0	1.0	1.0	0.0	19.67	1582.0	126.20	5.0	12.50	10.0	450.0
2	2.0	9.0	46000.0	3.0	1.0	0.0	18.20	1199.0	88.70	5.0	4.50	9.0	889.0
3	2.0	8.0	87000.0	1.0	1.0	0.0	20.77	1248.0	88.76	7.0	6.00	18.0	606.0
4	3.0	7.0	40670.0	1.0	0.0	2.0	15.20	1968.0	140.80	5.0	17.74	1.0	93.0

{ 'Brand' ,'Location', 'Fuel_Type', 'Transmission' ,'Owner_Type', 'Model', 'Engine', 'Power' }

Now we will slip our data into Test and Training

Data So, that we can test our DATA accuracy on a

different Models.

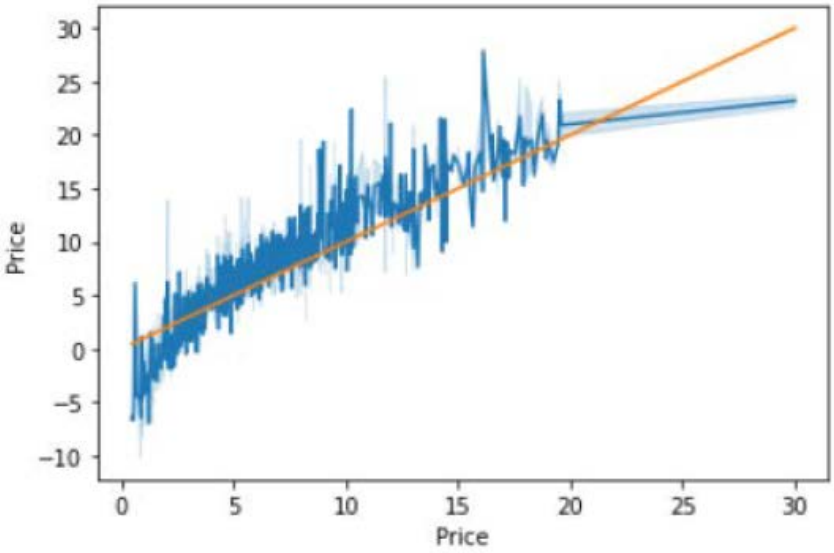
- Now we will split all The data into 2 parts x,y

  x : containing all the features

  y : containing Price

- Now, we pass or Data into Different Models

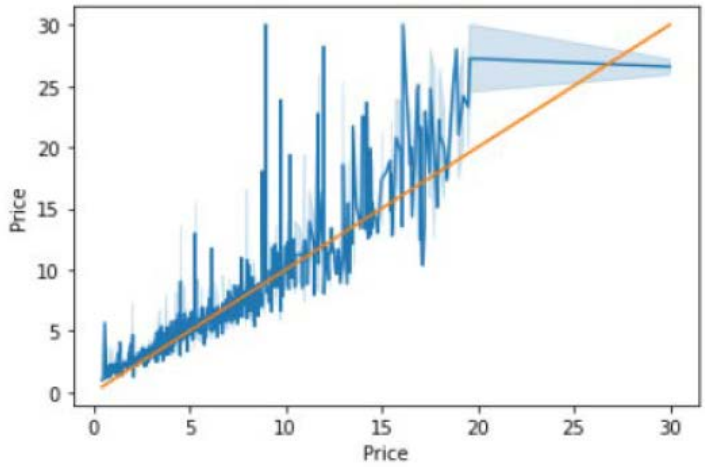
Linear Regression



Accuracy:
77.8822979113963%

Multiple linear regression (MLR), also known simply as multiple regression, is a statistical technique that uses several explanatory variables to predict the outcome of a response variable. The goal of multiple linear regression (MLR) is to model the linear relationship between the explanatory (independent) variables and response (dependent) variable.

KNeighborsRegressor

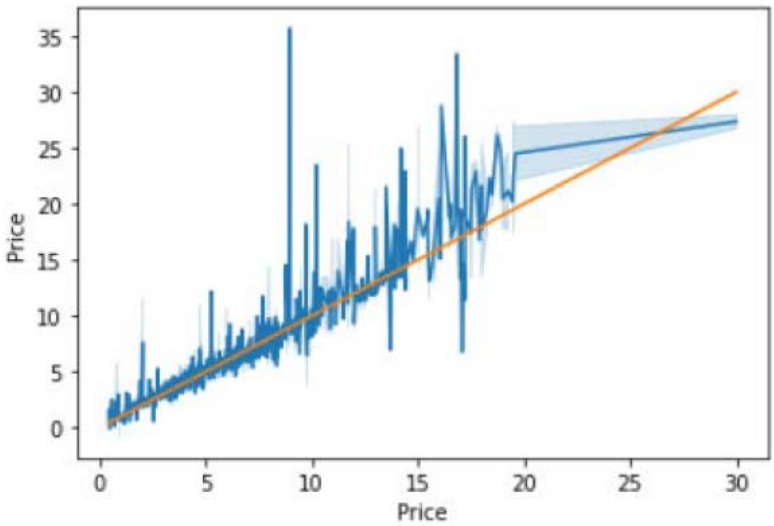


Accuracy:
88.38561409866529%

K-nearest neighbour is a machine learning technique in which the new (unknown) data is compared to all the existing records to locate the best match(es). Despite its apparent simplicity, a lot of taking has to be taken in preprocessing the data otherwise we can easily go off-track.

Both for classification and regression, a useful technique can be to assign weights to the contributions of the neighbors, so that the nearer neighbors contribute more to the average than the more distant ones. For example, a common weighting scheme consists in giving each neighbor a weight of 1/*d*, where *d* is the distance to the neighbor.

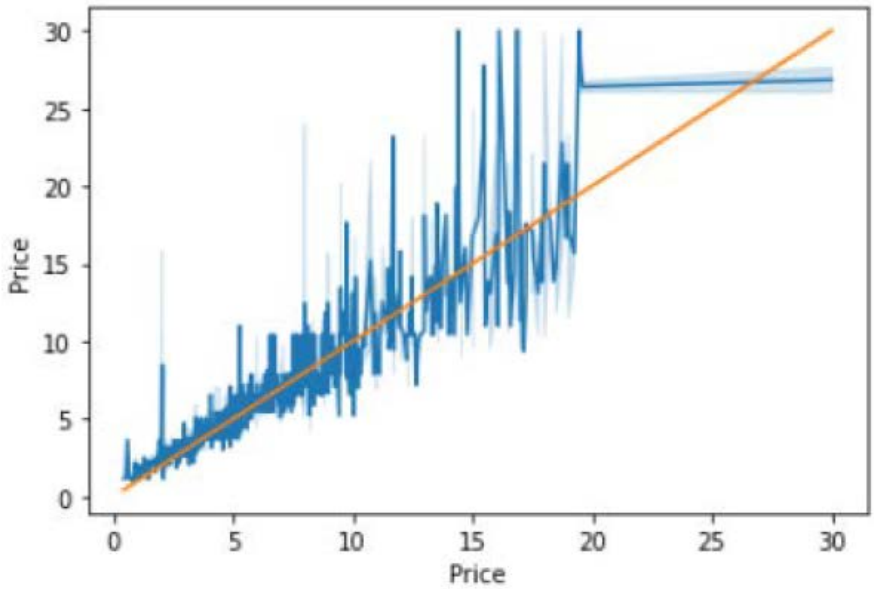
Support Vector Regression



Accuracy:
89.98166490143245%

"Support Vector Machine" (SVM) is a supervised machine learning algorithm that can be used for both classification or regression challenges. However, it is mostly used in classification problems. In the SVM algorithm, we plot each data item as a point in n-dimensional space (where n is the number of features you have) with the value of each feature being the value of a particular coordinate. Then, we perform classification by finding the hyper-plane that differentiates the two classes very well

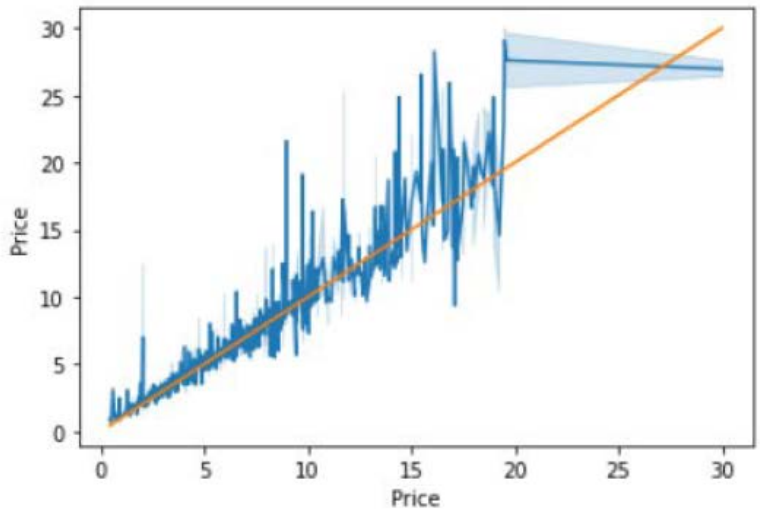
DecisionTreeRegressor



Accuracy:
84.47259346148451%

Decision tree builds regression or classification models in the form of a tree structure. It breaks down a dataset into smaller and smaller subsets while at the same time an associated decision tree is incrementally developed. The final result is a tree with **decision nodes** and **leaf nodes**.

RandomForestRegressor



We will use this model

Accuracy:
91.48808418027862%

Random forest algorithm can be used for both classifications and **regression** task. It provides higher accuracy through cross-validation. **Random forest** classifier will handle the missing values and maintain the accuracy of a large proportion of data.

Advantages and Disadvantages of Random Forest

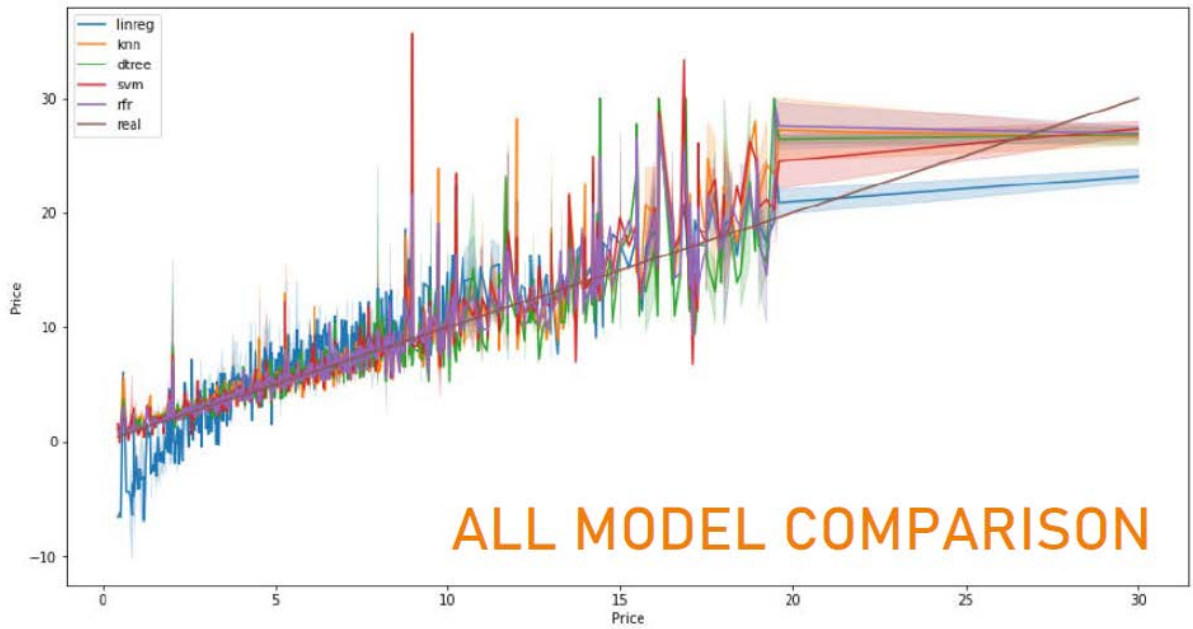
1. It reduces overfitting in decision trees and helps to improve the accuracy
2. It is flexible to both classification and regression problems
3. It works well with both categorical and continuous values
4. It automates missing values present in the data

However, despite these advantages, a random forest algorithm also has some drawbacks.

1. It requires much computational power as well as resources as it builds numerous trees to combine their outputs.
2. It also requires much time for training as it combines a lot of decision trees to determine the class.

Result and Conclusion

In this project we used five different machine learning techniques to forecast the price of used cars. After analysing all the techniques and their corresponding accuracies we concluded that random forest regressor had the best accuracy among all to predict the price of used car.



After selecting the model

- we pre-processed our test data similar to the training data.
- Then, we feed this data to our model and created Predict excel file in which we predicted the data.

```
predicted.to_excel (r'predicted_ans2.xlsx', index = True, header=True)
```

	A	B
1		**Price**
2	**0**	2.889619589
3	**1**	3.329460859
4	**2**	22.70620537
5	**3**	4.71446085
6	**4**	13.42858124
7	**5**	20.60961914
8	**6**	2.621676207
9	**7**	4.652359486
10	**8**	12.76007366
11	**9**	14.96189117
12	**10**	5.629328728
13	**11**	2.976163626
14	**12**	3.426496267
15	**13**	5.648015976
16	**14**	29.39456558
17	**15**	20.88275337
18	**16**	2.419197083
19	**17**	7.586062431
20	**18**	5.684495926
21	**19**	4.320672989
22	**20**	2.797524929
23	**21**	3.351922035
24	**22**	7.752715111
25	**23**	1.362505198
26	**24**	1.083096266
27	**25**	5.782477856
28	**26**	8.125685692
29	**27**	3.446851969
30	**28**	4.501909733

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